

Probabilistic infinitist Justification

Supplement for MCMP Poster Presentation

Clarification of the infinitist argument:

Here is a simple argument for infinitism using *the Principle of Avoiding Arbitrariness (PAA)* and *the Principle of Avoiding Circularity (PAC)*:

Premise 1 (PAA): For all x , if a person S has a justification for x , then there is some reason r_1 available to S for x ; and there is some reason r_2 available to S for r_1 ; etc., and there is no last reason in the series.¹

Premise 2 (PAC): For all x , if a person S has a justification for x , then for all r_i , if r_i is in the evidential ancestry of x for S , then x is not in the evidential ancestry of r_i for S .²

Infinitist Conclusion: For all x , if a person S has a justification for x , then x is justified by an infinite chain of non-repeating reasons.

For any of S 's justified beliefs x , PAA gives the following chain of reasons:

x (*justified by*) r_1 (*justified by*) r_2 (*justified by*) r_3 (*justified by*)...

This chain is infinite, so if one is justified, then further reasons are always available. The notion of availability is purposefully left rather vague.³ As long as it can be granted that such a notion of availability exists at all, we do not need to specify its exact strength for our purposes.

PAC states that the same belief cannot appear more than once within the same chain. This is assured by the notion of evidential ancestry, according to which all reasons within S 's justificatory chain must be distinct from the belief x ($x \neq r_i$ for all i), and all reasons within S 's justificatory chain must be distinct from each other ($r_i \neq r_j$ iff $i \neq j$). Since PAA entails that the chain of reasons which justifies S 's belief x is infinite, and PAC entails that all reasons within the chain are non-repeating, the infinitist conclusion follows from PAA and PAC.

¹ Peter Klein 1999, "Human Knowledge and the Infinite Regress of Reasons", 298-299 and Peter Klein 2005, "Infinitism is the Solution to the Regress Problem", 279.

² Peter Klein 1999, "Human Knowledge and the Infinite Regress of Reasons", 298-299. My notation.

³ Peter Klein 2005, "Infinitism is the Solution to the Regress Problem", 279-280.

Some technical details of the probabilistic infinitist model:

Under usual conditions, the infinite series determining $P(x)$ converges to a determinate positive limit. To show this, let's introduce shorthands for the involved terms and add indexes for the different conditional probabilities, since their values will of course differ from reason to reason:

$$\begin{aligned}x &= r_0 \\ \alpha_n &= P(r_n | r_{n+1}) \\ \beta_n &= P(r_n | \neg r_{n+1}) \\ \gamma_n &= \alpha_n - \beta_n\end{aligned}$$

With this notation and the Law of Total Probability in mind, the unconditional probability for x simplifies to:

$$\begin{aligned}P(r_n) &= \alpha_n P(r_{n+1}) + \beta_n P(\neg r_{n+1}) \\ &= \alpha_n P(r_{n+1}) + \beta_n (1 - P(r_{n+1})) \\ &= \alpha_n P(r_{n+1}) + \beta_n - \beta_n P(r_{n+1}) \\ &= \beta_n + \gamma_n P(r_{n+1})\end{aligned}\tag{1}$$

The infinite justificatory chain does not need a foundational proposition. But to show that such a proposition is not needed, let's call r_f the foundational proposition. A chain with the foundational proposition (for any $f \geq 0$) looks like this⁴:

$$P(r_0) = \beta_0 + \gamma_0 \beta_1 + \gamma_0 \gamma_1 \beta_2 + \dots + \gamma_0 \gamma_1 \dots \gamma_{f-2} \beta_{f-1} + \gamma_0 \gamma_1 \dots \gamma_{f-1} P(r_f)\tag{2}$$

We now need to understand what happens when we introduce additional reasons. What happens is that the value of the foundational proposition r_f lessens when more reasons are included in the chain. This is because every reason we add also adds another γ_n term (consisting of the two relevant conditional probabilities). Since $0 < \gamma_n < 1$ for all γ_n , and since r_f is multiplied by all of the γ_n terms, its value decreases for every additional reason and goes to 0 in the limit. What remains is a series which converges to a well-defined justificatory value for x , entirely based on the conditional probabilities:

$$P(r_0) = \beta_0 + \gamma_0 \beta_1 + \gamma_0 \gamma_1 \beta_2 + \gamma_0 \gamma_1 \gamma_2 \beta_3 + \dots\tag{3}$$

Crucially, this means that as the chain of reasons becomes sufficiently long, the foundational proposition becomes entirely irrelevant, which proves that foundational propositions are not necessary for an infinite regress of probabilities to have a well-defined value. This

⁴ proof is given in A&P Appendix A1

phenomenon is called 'Fading Foundations' by A&P: "the effect of the foundational term in a finite chain diminishes as the chain becomes longer, and completely dies away in the limit".⁵ Note that there are some additional minor constraints that the chain of reasons must fulfill in order for Fading Foundations to occur.⁶ Regardless, it has been shown an infinite regress of probabilities can give a well-defined justificatory value. Due to this formal model, the concept of probabilistic infinitism is shown to be coherent at the very least.

About accuracy:

The accuracy of our judgment of the justification of x is determined by the range of possible values that $P(x)$ can take with a given chain of reasons.

Say we have only taken a finite amount of reasons into account in our calculation of $P(x)$.⁷ Must this mean that we are *unjustified* in x according to probabilistic infinitism? Not necessarily. But our judgment of $P(x)$ will not be perfectly accurate, because our chain will include a foundational reason $P(r_i)$, and the value of this proposition is unknown to us because we do not calculate its value using the preceding reason and the Law of Total Probability. Since $0 \leq P(r_i) \leq 1$, we can still get a more or less accurate judgment of $P(x)$ by calculating $P(x)$ when $P(r_i)=1$, and then calculating $P(x)$ when $P(r_i)=0$. The true value of $P(x)$ must now fall within the range of these two calculated values. Since the impact of $P(r_i)$ on $P(x)$ lessens as more reasons are added (due to Fading Foundations, see the previous section), the range of possible values for $P(x)$ also lessens as more reasons are added, and we thus get a more accurate estimate of $P(x)$.

⁵ A&P, 83.

⁶ A&P ch. 3.6

⁷ Infinitely many reasons must perhaps be *available* in a sense, but we have not actively used an infinite amount of reasons in our justification of x .